Michael Adu

Embedded Software Engineer

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PROFILES

Github

Michael-Adu

Linkedin

Michael Adu

SUMMARY

A passionate, motivated and project-oriented Embedded Systems Engineer, constantly looking for a challenge and the opportunity to further my knowledge and expand my skills. Skilled in embedded programming languages such as C/C++and have extensive experience with microcontrollers and SoCs such as Arduinos, Raspberry Pi's, ARM Cortex-M microcontrollers and PIC18 microcontrollers.

EDUCATION

Coventry University — Masters • Distinction

Embedded Systems Engineering — Coventry, United Kingdom • 2022 - 2023

Ashesi University — Bachelors

Electrical and Electronic Engineering — Berekuso, Ghana • 2017-2021

EXPERIENCE

Skyships Automotive Ltd. — Braintree, United Kingdom

Software Engineer — January 2024 - Present

Graduate Embedded Software Engineer — January 2024 - November 2025

- Designed and developed automotive infotainment and cockpit controllers for high end vehicles, integrating CAN messages with various telltales and warnings using QNX.
- Created several tools to aid in software testing, including a suite of tools to exponentially reduce the duration necessary for calculating and sending automotive CAN messages, performing various critical tests.
- Developed 2D and 3D user interfaces using Kanzi and Lua.

Premium Technologies Ltd. — Accra, Ghana

Embedded Engineer and Lead Web Developer — February 2022 - August 2022

- Designed and developed a custom Smart Home System using ESP8022 to interface with already established home electrical systems to be implemented in over 3000 homes.
- Created a Linux-based Smart Home Hub running on a Raspberry Pi and a companion phone application using Flutter to enable user control of household appliances and monitoring of CCTV cameras on a local network.

Mingo Blox — Accra, Ghana

Embedded Engineer and Web Developer — June 2021 - February 2022

- Developed a desktop and mobile graphical-based code application for quick and easy programming of Arduino devices for novel engineering students and hobbyists using ReactJS and C/C++, which was marketed to an estimated 30 million users
- Taught the fundamentals of robotics to 30+ young students at the primary level

PROJECTS

Automated Sign Language Detection Device 2022-2023

Developing a physical device to recognize sign language hand gestures with the aid of a PIC18F, piezoelectric sensors and machine learning to aid potentially over 430 million audible-impaired individuals with communication.

Trained a Long Short Term Model Neural Network model to recognize gestures made by the user. Skills used: C/C++, Proteus, Python, PIC18F development, Git and MATLAB.

Yocto GUI 2022-Present

Developing a graphical user interface for creating Linux Yocto-based operating systems for embedded systems.

The application aims to bring ease of access to Yocto and provide a centralised workflow when developing operating systems to aid in an industry with an estimated $116 billion market size.
Skills used: C/C++, Flutter, NodeJS, Linux

Range Sensor 2022

Developed a custom Embedded Linux operating system using Yocto to be deployed on a Raspberry Pi with a custom Ultrasonic Sensor device Driver

Skills used: Yocto, BitBake, C/C++

SKILLS

Coding

C/C++, Python, VHDL, HTML, Javascript, Dart, PHP, SQL, Lua

Microcontrollers/SoC

Qualcomm, PIC18, STM32, Arduino, Raspberry Pi, ESP32

Communication Protocols

CAN, UART, UDP/TCP, SPI, I2C

Operating Systems

Linux, QNX, RTOS

Code Project Management

Git, Github

Application Frameworks

React, Flutter, QT, PyQT

HMI

Kanzi

Others

Blender, Figma

INTERESTS

3D Modelling

REFERENCES

Available Upon Request